

SmartCare v0.5 - Architecture and Refactoring Workbook

Week 8 student resource

Current Architecture Problems

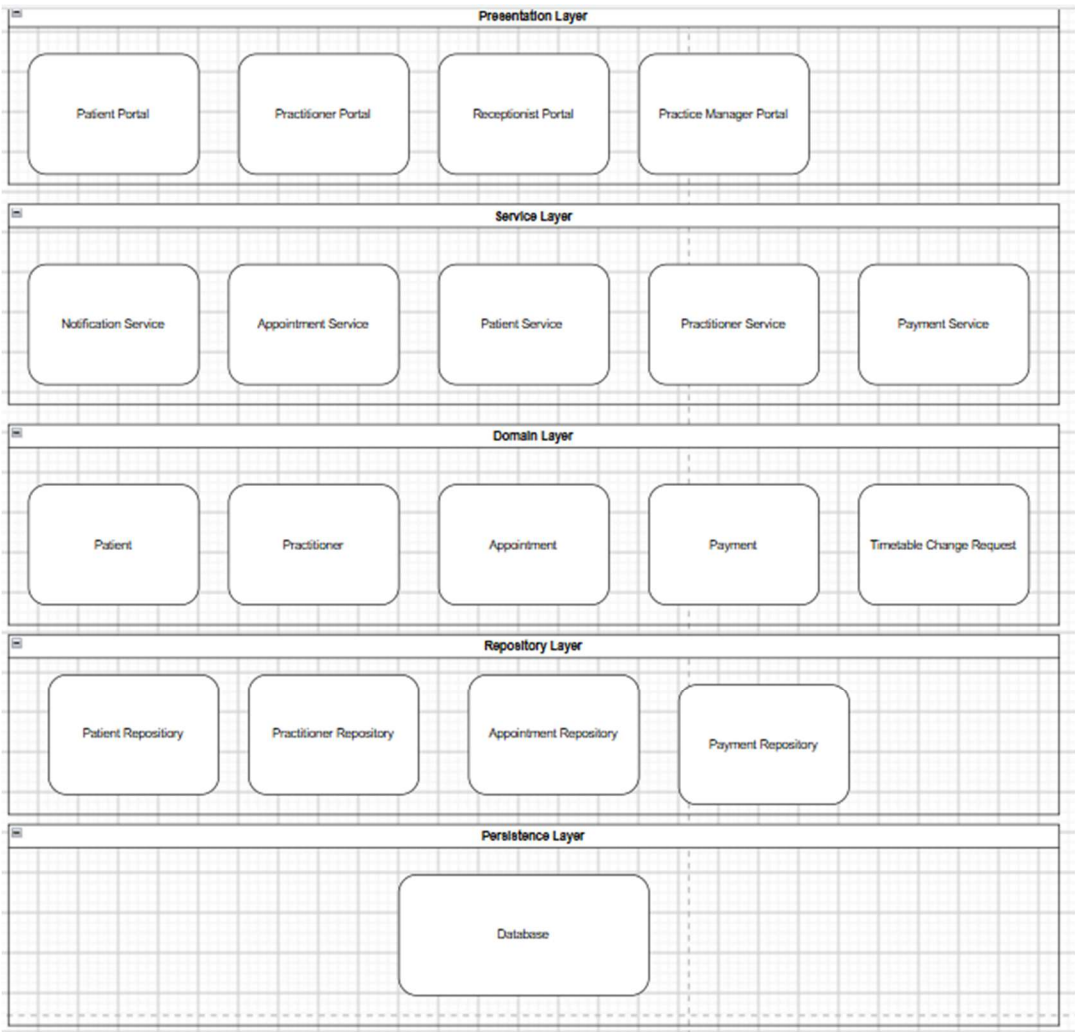
Problem	Evidence	Impact	Refactoring
Syntax errors	The previous code had multiple syntax errors and could use validation	The software produces incorrect results or behaves unpredictably because the logic does not match the intended workflow.	Correct all syntax mistakes, run code validation tools, and ensure the program compiles before adding new features.
Logical errors	The code did not operate using typical logic code	The software produces incorrect results or behaves unpredictably because the logic does not match the intended workflow.	Rewrite the logic to follow correct control flow, add conditional checks, and test each function to confirm expected behaviour.
The database is already in the database.	The database will not be uploaded properly because a database will already exist. This can be shown with the new statement that will be added: IF NOT EXISTS.	The system may crash or fail to initialise because it attempts to recreate an existing database, causing duplication or errors.	Add IF NOT EXISTS to database creation statements and implement checks to avoid recreating existing structures.
Indentation	In order to make it clearer to understand, indentation should be properly observed; this can be reviewed with regard to past code.	Poor readability makes the code harder to maintain, debug, or extend, increasing the chance of future mistakes.	Apply consistent indentation and formatting standards across all files to improve clarity and maintainability.
Commenting	Commenting was not shown in previous code, making it hard to understand the direction the software was meant to take.	Without comments, it becomes difficult to understand the purpose of functions or the intended workflow, slowing down development.	Add clear comments about function roles to guide future maintenance and collaboration.

Layer Responsibilities

Layer	Responsibilities	Must not contain
Presentation	Will include Appointments and any viewable patient records.	SQL queries, business logic, validation rules beyond simple UI checks.
Service	Fetches SQL table IDs and validates them.	UI code, database schema details, raw SQL statements.

Architecture Diagram

Insert SmartCare v0.5 architecture and dependency direction.



SOLID Review

Principle	Relevant?	Evidence	Decision
SRP			
OCP			
LSP			
ISP			
DIP			

AI Architecture Review

AI suggestion	Observed problem?	Decision	Reason	Verification

